

FIVE-LANE COST RESULTS FOR RECONSTRUCTABLE GOVERNANCE

Observed execution, normalized admissibility, and bounded economic interpretation

Experiment	SV-COST-FIVE-LANE-RESULTS-001
Task	SV-RECON-001
Operation class	governed_state_reconstruction
Comparison unit	successful equivalent admissible outcome
Publication status	RESULTS_READY_FOR_BOUNDED_PUBLICATION

All five lanes completed successfully, preserved task identity, produced the same normalized admissible outcome, and passed the publication gate. This does not establish universal provider economics, company-wide ROI, or fresh-inference equivalence.

Headline result

For this bounded reconstruction task, StegVerse-only deterministic reconstruction was the lowest-cost successful equivalent admissible lane. OpenAI governance was approximately cost-neutral (+0.073%), while Anthropic governance was 33.22% lower than Anthropic raw in this observed run.

Lane	Cost / admissible outcome	Latency	Result
OpenAI raw	\$0.006875	3.424027735 s	PASS
OpenAI governed	\$0.006880	2.867607194 s	PASS
Anthropic raw	\$0.010656	7.939445812 s	PASS
Anthropic governed	\$0.007116	5.459360124 s	PASS
StegVerse-only deterministic reconstruction	\$0.000000002885	0.000000461 s	PASS

Provider costs are reconstructed from retained token-usage receipts using a versioned declared price card and are not invoice-reconciled charges. The StegVerse figure combines measured runtime and output size with declared runner and storage rates.

1. Research question and task contract

Can OpenAI raw, OpenAI governed, Anthropic raw, Anthropic governed, and StegVerse-only deterministic reconstruction produce the same successful equivalent admissible outcome under one task contract, and what is the declared-rate cost of each lane?

The task began with balance 100, risk score 1, and active standing. Six ordered events were evaluated. Credits increase balance; debits are denied when they cross minimum balance or standing is not active; risk additions are denied when they exceed the maximum risk score; denied events never mutate state.

Event	Operation	Amount	Required decision
E01	Credit	25	ALLOW - CREDIT_APPLIED
E02	Debit	40	ALLOW - DEBIT_WITHIN_BOUNDARY
E03	Risk add	2	ALLOW - RISK_WITHIN_BOUNDARY
E04	Debit	100	DENY - MINIMUM_BALANCE_VIOLATION
E05	Risk add	3	DENY - MAXIMUM_RISK_VIOLATION
E06	Debit	10	ALLOW - DEBIT_WITHIN_BOUNDARY

Required final state: balance 75; risk score 3; standing active; applied events 4; denied events 2.

2. Admissibility and normalization

Provider outputs were not admitted merely because they were valid JSON or semantically plausible. Each output was normalized into the same contract and compared against task identity, final state, ordered event decisions, applied count, denied count, and claim boundary. Missing, contradictory, unsupported, or incorrect transitions failed closed. All five passing lanes produced the same normalized outcome hash.

sha256:155869baef4bd023ad95e63c6a81d6ade921e92660cec351680e1aabd4d2597

3. How the cost numbers were generated

$$\text{Provider cost} = (\text{input tokens} \times \text{input rate} + \text{output tokens} \times \text{output rate}) / 1,000,000.$$

Declared rates: OpenAI input \$5.00 per million and output \$30.00 per million; Anthropic input \$3.00 per million and output \$15.00 per million.

$$\text{StegVerse local cost} = \text{runtime seconds} \times (\$0.008 / 60) + \text{output GB} \times \$0.008.$$

Measured StegVerse-only runtime was approximately 0.000000461 seconds and normalized output size was 353 bytes. The modeled compute component was approximately \$0.000000000061 and storage component approximately \$0.000000002824, for a total of \$0.000000002885.

4. Provider-pair findings

Provider	Raw	Governed	Delta	Observed delta
OpenAI	\$0.006875	\$0.006880	+\$0.000005	+0.072727%
Anthropic	\$0.010656	\$0.007116	-\$0.003540	-33.220721%

OpenAI governance was effectively cost-neutral in this run. Anthropic governance used fewer output tokens and was 33.22% lower than Anthropic raw while producing the same normalized admissible outcome. This is a single-run observation, not a guaranteed provider effect.

5. Bounded reconstruction comparison

StegVerse-only did not perform fresh open-ended model inference. It deterministically reconstructed an already-defined admissible successor state from the task contract. Because all five lanes produced the same normalized admissible result for this operation, a matched-operation comparison is valid here.

Provider-to-StegVerse cost ratios ranged from approximately 2.38 million to 3.69 million times, with matched-operation modeled reductions above 99.99995% for this bounded deterministic reconstruction.

These very large ratios must not be generalized to discovery, open-ended reasoning, fresh inference, all enterprise workloads, or company-wide ROI. They identify where deterministic reconstruction can avoid unnecessary new generation; they do not establish that deterministic reconstruction replaces frontier-model capability.

6. What the result establishes

- A provider-neutral five-lane harness can enforce admissibility before cost selection.
- All five lanes can produce the same normalized governed-state result under one contract.
- Provider-call governance overhead can be close to zero or negative depending on output behavior.
- Deterministic reconstruction can be materially cheaper when reconstruction is the correct operation.
- Cost should be reported per successful equivalent admissible outcome, not merely per attempt or token count.

7. What the result does not establish

- Replacement of foundation models for fresh reasoning or discovery.
- Universal savings across providers, workloads, or organizations.
- Invoice-reconciled provider charges or fully burdened enterprise cost.
- Company-wide ROI or fresh-inference equivalence.

8. Conclusion and evidence

The five-lane experiment completed with all lanes successful, equivalent, and admissible. StegVerse-only deterministic reconstruction was the lowest-cost admissible lane for this bounded operation. The evidence supports a narrow conclusion: reconstructable governance can be technically available, provider-neutral, admissibility-preserving, and economically negligible or advantageous for bounded state reconstruction.

Broader provider-profit, enterprise ROI, and fresh-inference claims require separate held-out tasks, repeated trials, invoice reconciliation, and fully burdened operating-cost analysis.

Evidence repository	GCAT-BCAT-Engine/workflows
Evidence commit	3720211a1cfaaf2db697f3e26194d083db21e94f
Publisher source	papers/five-lane-reconstructable-governance-analysis.md
Site source	papers/sv-cost-relational-analysis.html
Task-contract hash	sha256:2e9b4a4193669b6d8f1d3fea8639d2adcee6090c58246b8b99920ba2f08dfb6b
Outcome hash	sha256:155869baaef4bd023ad95e63c6a81d6ade921e92660cec351680e1aabd4d2597
Price-card status	VERSIONED_DECLARED_RATE_NOT_INVOICE_RECONCILED

Public paper: <https://stegverse.org/papers/sv-cost-relational-analysis.html>

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